ARTIFICIAL INTELLIGENCE – CAT 1

Total Marks: 15

Instructions: Answer all questions. Where appropriate, support your answers with relevant examples.

Question 1: (5 Marks)

a) Define **Artificial Intelligence (AI)** and explain **two characteristics** that distinguish an AI system from a conventional computer program. **(3 marks)**
b) Identify **two real-world applications of AI** and explain how AI is used in each application. **(2 marks)**

Question 2: (5 Marks)

A university develops an AI-based student-support system that receives questions from students, analyses them, and provides appropriate responses.

a) Define an **intelligent agent**. **(2 marks)**
b) Identify and explain **three components** involved in the operation of an intelligent agent. **(3 marks)**

Question 3: (5 Marks)

A delivery company wants to develop an AI system that determines the shortest route for delivering packages to different locations.

a) Explain how **problem formulation** can be used to represent this delivery problem. **(2 marks)**
b) Explain the role of **search algorithms** in solving AI problems, giving **one example of a search algorithm**. **(2 marks)**
c) State **one limitation or challenge** that may arise when using AI to solve real-world problems. **(1 mark)**